

8.5. 傅里叶变换应用于通讯系统

一、信号通过系统的频域分析方法

1. $H(j\omega) = \frac{Y(j\omega)}{F(j\omega)}$ 响应的频谱 = $H(j\omega) e^{j\omega t}$
 幅频改变: $|Y(j\omega)| = |H(j\omega)| \cdot |F(j\omega)|$
 相位改变: $\angle Y(j\omega) = \angle H(j\omega) + \angle F(j\omega)$

2. $H(j\omega)$ 的成立条件满足:

- ① $h(t)$ 绝对可积
- ② $H(j\omega)$ 的初值均位于虚平面上

3. 周期信号激励时的响应

输入信号为余弦信号即

$f(t) = A \cos(\omega t)$
 $F(j\omega) = \pi A [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$

$Y(j\omega) = H(j\omega) F(j\omega)$

$= \pi A H(j\omega) [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$

$= \pi A [H(-j\omega_0) \delta(\omega + \omega_0) + H(j\omega_0) \delta(\omega - \omega_0)]$

$= \pi A |H(j\omega_0)| [e^{-j\phi_0} \delta(\omega + \omega_0) + e^{j\phi_0} \delta(\omega - \omega_0)]$

$y(t) = A |H(j\omega_0)| \left(\frac{e^{-j\phi_0} e^{j(\omega_0 t + \omega_0 t)} + e^{j\phi_0} e^{j(\omega_0 t - \omega_0 t)} \right)$

$= |H(j\omega_0)| A \cos(\omega_0 t + \phi_0)$

4. 非正弦周期信号激励时的响应

$f_n = H(jn\omega) F_n$ 为输出信号的频谱

$Y(j\omega) = F(j\omega) H(j\omega)$

$y(t) = F^{-1}[Y(j\omega)]$

二、系统无失真传输的条件

1. 失真: 系统的响应波形与激励波形不相同

无失真: 响应信号与激励信号相比, 只是大小与出现时间不同, 而波形不变

2. 无失真传输的条件: $y(t) = K f(t - t_0)$, t_0 为滞后时间

$f(t) \rightarrow h(t) = K \delta(t - t_0) \rightarrow y(t) = f(t) * h(t)$

$Y(j\omega) = F[K \delta(t - t_0)] F(j\omega) = K F(j\omega) e^{-j\omega t_0}$

$H(j\omega) = K e^{-j\omega t_0}$

3. 相位失真的条件:

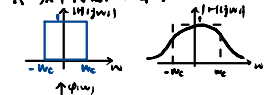
input: $f(t) = A_1 \sin(\omega_1 t) + A_2 \sin(2\omega_1 t)$

output: $y(t) = K A_1 \sin(\omega_1 t - \phi_1) + K A_2 \sin(2\omega_1 t - \phi_2)$

不产生相位失真: $\frac{\phi_1}{\omega_1} = \frac{\phi_2}{2\omega_1} = t_0$ 即 $\frac{\phi_1}{\phi_2} = \frac{1}{2}$

三、理想低通滤波器

1. 频率特性: $H_L(j\omega) = G \omega_c \text{rect}(\frac{\omega}{2\omega_c})$



$H_L(j\omega) = \begin{cases} e^{-j\omega t_0} & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$

2. 理想低通滤波器的冲激响应

$h(t) \leftrightarrow H_L(j\omega)$

$h(t) = F^{-1}[H_L(j\omega)] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H_L(j\omega) e^{j\omega t} d\omega$

$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{-j\omega t_0} e^{j\omega t} d\omega$

$= \frac{1}{\pi} \int_{-t_0}^{t_0} \frac{1}{\omega} [e^{j\omega(t-t_0)} - e^{-j\omega(t-t_0)}] d\omega$

$h(t) = \frac{\omega_c}{\pi} \frac{\sin[\omega_c(t-t_0)]}{\omega_c(t-t_0)}$
 $= \frac{\omega_c}{\pi} S_a[\omega_c(t-t_0)]$

① 比较输入输出, 可见严重失真

$\delta(t) \leftrightarrow 1$ 信号带宽无限

而理想低通的通频带(系统带宽)有限的 $[0, \omega_c]$

$\delta(t)$ 经过理想低通时, ω_c 以上的频率成分都衰减

为 0 $\rightarrow$ 失真

$\omega_c \rightarrow \infty$ $h(t) \rightarrow \delta(t)$

系统为全通网络, 可以无失真传输

② 理想低通滤波器是个物理不可实现的非因果系统

$t < 0$ 有非零的冲激响应

3. 理想低通的阶跃响应

激励: $e(t) = u(t) \leftrightarrow \pi \delta(\omega) + \frac{1}{j\omega}$

系统: $h(t) \leftrightarrow H_L(j\omega) = \begin{cases} e^{-j\omega t_0} & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$

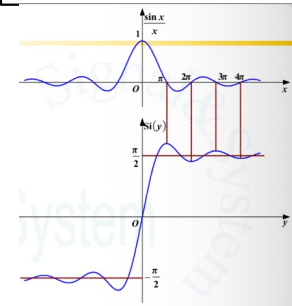
响应: $R_L(\omega) = u(\omega) * h(\omega) = \begin{cases} e^{-j\omega t_0} & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$

$r(t) = F^{-1}[R_L(\omega)]$

$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} [\pi \delta(\omega) + \frac{1}{j\omega}] e^{-j\omega t_0} e^{j\omega t} d\omega$
 $= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \pi \delta(\omega) e^{j\omega(t-t_0)} d\omega + \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{1}{j\omega} e^{j\omega(t-t_0)} d\omega$
 $= \frac{1}{2} + \frac{1}{\pi} \int_0^{\omega_c(t-t_0)} \frac{\sin x}{x} dx$

正弦积分: $\text{Si}(y) = \int_0^y \frac{\sin x}{x} dx$

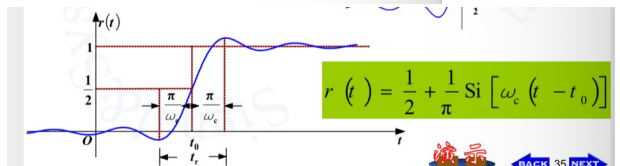
- ① 下限 0
- ② 奇函数
- ③ $\max: x = \infty \quad \lim_{x \rightarrow \infty} \text{Si}(x) = \frac{\pi}{2}$



① 上升时间 t_r : 输出由 $\min \rightarrow \max$ 时间

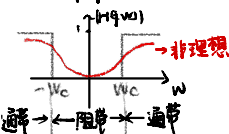
$t_r = \frac{2\pi}{\omega_c} = \frac{1}{B}, B = \frac{\omega_c}{2\pi} = f_c$

② $B t_r = 1$



4. 理想高通滤波器的频率特性

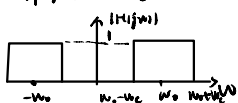
$$H_{HP}(j\omega) = 1 - G_{2\omega_c}(\omega)$$



通带 ← 阻带 → 通带

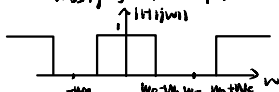
5. 理想带通滤波器的频率特性

$$H_{BP}(j\omega) = G_{2\omega_c}(\omega - \omega_0) + G_{2\omega_c}(\omega - \omega_0)$$



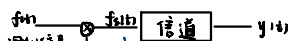
6. 理想带阻滤波器的频率特性

$$H_{BS}(j\omega) = 1 - H_{BP}(j\omega)$$



四. 调制与解调

1. 调制: 把各种信号的频谱搬移, 使它们互不重叠地居于不同频率范围



$$y(t) = f(t) \cos \omega_c t$$

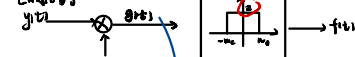
$$f_s(t) = f(t) \cos \omega_c t$$

$$f_s(t) = f(t) \cos \omega_c t$$

$$F_s(j\omega) = \frac{1}{2} [F(j\omega - \omega_c) + F(j\omega + \omega_c)]$$

2. 解调

已知信号 $y(t) = f(t) \cos \omega_c t$



$$f(t) = y(t) \cos \omega_c t = f(t) \cos^2 \omega_c t = \frac{1}{2} [f(t) + f(t) \cos 2\omega_c t]$$

$$f(t) = \frac{1}{2} [f(t) + f(t) \cos 2\omega_c t]$$

$$F(j\omega) = \frac{1}{2} [F(j\omega) + \frac{1}{2} (F(j\omega - 2\omega_c) + F(j\omega + 2\omega_c))]$$

为信号的频谱通过理想低通滤波器, 可取出 $F(j\omega)$, 从而恢复原信号 $f(t)$

五. 带通滤波系统的应用

六. 从抽样信号恢复连续信号



1. 时域运算:

$$f_s(t) = f(t) \delta(t - nT_s)$$

$$F_s(j\omega) = \frac{1}{T_s} F(j\omega) * \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_c) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(j\omega - n\omega_c)$$

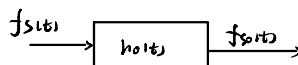
$$H(j\omega) = \begin{cases} T_s & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \leftrightarrow h(t) = T_s \frac{\omega_c}{\pi} \text{Sa}(\omega_c t)$$

$$f(t) = f_s(t) * h(t) = \left[\sum_{n=-\infty}^{\infty} f(nT_s) \delta(t - nT_s) \right] * \left[T_s \frac{\omega_c}{\pi} \text{Sa}(\omega_c t) \right]$$

$$= \frac{T_s \omega_c}{\pi} \sum_{n=-\infty}^{\infty} f(nT_s) \text{Sa}(\omega_c (t - nT_s))$$

2. 零阶抽样保持

$$h(t) = u(t) - u(t - T_s) = G_{T_s}(t - \frac{T_s}{2}) \leftrightarrow H(j\omega) = T_s \text{Sa}(\frac{\omega T_s}{2}) e^{-j\omega \frac{T_s}{2}}$$



$$f_s(t) = f(t) * h(t)$$

$$F_s(j\omega) = F(j\omega) H(j\omega)$$

$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(j\omega - n\omega_c) [T_s \text{Sa}(\frac{\omega T_s}{2}) e^{-j\omega \frac{T_s}{2}}]$$

$$= \sum_{n=-\infty}^{\infty} F(j\omega - n\omega_c) \text{Sa}(\frac{\omega T_s}{2}) e^{-j\omega \frac{T_s}{2}}$$

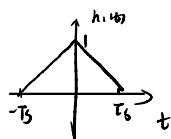
系统滤波器 $H_0(j\omega)$

$$H_0(j\omega) = \begin{cases} 1 \cdot e^{-j\omega \frac{T_s}{2}} & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

$$\text{允许延迟 } t_d = \frac{T_s}{2}$$

$$|H_0(j\omega)| = \frac{1}{\text{Sa}(\frac{\omega T_s}{2})} \quad |\omega| < \omega_c$$

3. 一阶抽样保持



$$h_1(t) = \begin{cases} 1 - \frac{|t|}{T_s} & |t| \leq T_s \\ 0 & |t| > T_s \end{cases}$$

$$F[h_1(t)] = T_s \text{Sa}^2(\frac{\omega T_s}{2})$$

$$F_s(j\omega) = F[f_s(t)]$$

$$= F[h_1(t)] \cdot F_s(j\omega)$$

$$= \sum_{n=-\infty}^{\infty} F(j\omega - n\omega_c) \text{Sa}^2(\frac{\omega T_s}{2})$$

$$H_1(j\omega) = \begin{cases} \frac{1}{\text{Sa}^2(\frac{\omega T_s}{2})} & |\omega| \leq \frac{\omega_c}{2} \\ 0 & |\omega| > \frac{\omega_c}{2} \end{cases}$$